Lab 2 Report: ROS, TF and Quaternions

Xuanye Xu

Student ID 24020036071 / H00490299

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[PENDING: summary written after all parts are complete]

Grading Map

Grading item	Points	Section	Evidence
ROS installation	20	Section 2	Environment table, turtlesim screenshot, node and topic listing
Two-drone motion	40	Section 3	Answers on nodes, topics and the launch file; publisher code; static and dynamic scenes; TF tree; `/tf` rate
Trajectories	30	Section 4	Three trajectories in the `world` and `av1` frames; numerical check against the analytic model
Mathematical derivation	10	Section 5	Five derivations with homogeneous transforms
Quaternion properties	10	Section 6	Three proofs; note on a sign difference in the lecture slides

1 *Lab Introduction*

This lab introduces the Robot Operating System (ROS 1 Noetic) and the `tf2` transform library. Two simulated drones are driven by publishing their poses as time-varying transforms, their trajectories are drawn in RViz in different reference frames, and the observed relative motion is explained with homogeneous transforms. The last part proves several properties of the quaternion multiplication matrices. The tasks follow the MIT VNAV 2023 Lab 2 exercises,[1] using the ROS 1 template code from the `VNAV2023` release.[2] The working environment is summarised in Table 1.

[1] MIT SPARK Lab. VNAV 2023 Lab 2: ROS Exercises. https://vnav.mit.edu/labs_2023/lab2/exercises.html, 2023a

[2] MIT SPARK Lab. VNAV-labs, release VNAV2023 (commit 7d234d8). https://github.com/MIT-SPARK/VNAV-labs/releases/tag/VNAV2023, 2023b

2 *ROS Installation* [20 pts]

Platform. All ROS work runs in Ubuntu 20.04 under WSL 2 on an x86-64 Windows 11 desktop (Table 1). Commands are issued from a MacBook over SSH; the Windows desktop is viewed with Moonlight, and Linux GUI programs (turtlesim, RViz) are displayed on it through WSLg.

Apple-silicon Macs cannot run the x86-64 simulators required in Labs 3 and 4, so the whole course uses this x86-64 host.

Installation. The ROS key used by older installation guides expired on 1 June 2025, so the package source was configured with the official `ros-apt-source` package (version 1.3.0), which installs the current signing key and the `packages.ros.org` source.[3] The desktop-full metapackage and the build tools were then installed with `apt`: 1487 new packages, including 234 `ros-noetic-*` packages, with no apt errors or warnings. Finally `rosdep` was initialised and updated with `-include-eol-distros`, because Noetic reached end of life in

[3] Open Robotics. ROS signing key migration guide. https://discourse.openrobotics.org/t/ros-signing-key-migration-guide/43937, 2025

Component	Version
Windows host	10.0.26200 (Windows 11), WSL 2.7.12, WSLg 1.0.73
Linux	Ubuntu 20.04.6 LTS, kernel 6.18.33.2-microsoft-standard-WSL2, x86_64
ROS	Noetic (`ros-noetic-desktop-full` 1.5.0)
Key packages	roscpp 1.17.4, tf2_ros 0.7.10, rviz 1.14.26, turtlesim 0.10.3
Build tools	catkin_tools 0.9.4, g++ 9.4.0, CMake 3.16.3
Lab code	VNAV-labs tag `VNAV2023`, commit `7d234d8`

Table 1: Software environment, taken from `results/logs/environment.txt`.

May 2025, and `source /opt/ros/noetic/setup.bash` was added to `~/.bashrc`.

Full command list and apt summary: `results/logs/ros_install_summary.txt`.

Turtlesim test. `roscore`, `turtlesim_node` and `turtle_teleop_key` were started in three shells. The teleoperation node reads the arrow keys and publishes `geometry_msgs/Twist` messages on `/turtle1/cmd_vel`, which the simulator subscribes to (Listing 1). While the turtle was driven by hand, all 32 published commands were recorded: 12 forward and 2 backward ($v_x = \pm 2$), 7 left and 11 right turns ($\omega_z = \pm 2$). Figure 1 shows the resulting path.

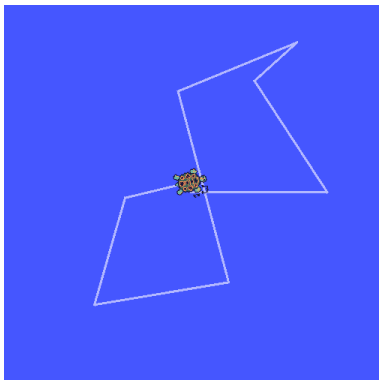


Figure 1: Turtlesim window after keyboard teleoperation; the trail is the path driven with the arrow keys.

Listing 1: Nodes and the command topic during teleoperation (`results/logs/turtlesim_nodes_topics.txt`).

```
--- rosnode list
/rosout
/teleop_turtle
/turtlesim
--- rostopic info /turtle1/cmd_vel
Type: geometry_msgs/Twist

Publishers:
 * /teleop_turtle (http://DESKTOP-91LK43L:44136/)

Subscribers:
 * /turtlesim (http://DESKTOP-91LK43L:44040/)
```

> EVIDENCE Table 1 (versions); `ros_install_summary.txt` (installation); Figure 1 (turtlesim with keyboard-driven path); Listing 1 (nodes and topic connections); `turtle_cmd_vel.csv` (32 recorded keyboard commands).

3 *ROS Basics and Two-Drone Motion* [40 pts]

3.1 *Nodes, topics and the launch file*

[PENDING: answers to the four questions, checked against the running system]

3.2 *Publishing the drone frames*

[PENDING: implementation of `frames_publisher_node`]

3.3 *Results*

[PENDING: static and dynamic scenes, TF tree, `/tf` rate]

4 *Trajectories and Reference Frames* [30 pts]

4.1 *Querying the relative transform*

[PENDING: `lookupTransform` direction and error handling]

4.2 *Trajectories in RViz*

[PENDING: screenshots with fixed frame `world` and `av1`]

4.3 *Numerical check*

[PENDING: sampled TF against the analytic relative position]

5 *Mathematical Derivations* [10 pts]

[PENDING: five derivations]

6 *Quaternion Properties* [10 pts]

[PENDING: three proofs and the note on the slides]

7 *Discussion*

[PENDING: problems met and how they were solved; reflections]

8 *Conclusions*

[PENDING:]

References

MIT SPARK Lab. VNAV 2023 Lab 2: ROS Exercises. https://vnav.mit.edu/labs_2023/lab2/exercises.html, 2023a.

MIT SPARK Lab. VNAV-labs, release VNAV2023 (commit 7d234d8). https://github.com/MIT-SPARK/VNAV-labs/releases/tag/VNAV2023, 2023b.

Open Robotics. ROS signing key migration guide. https://discourse.openrobotics.org/t/ros-signing-key-migration-guide/43937, 2025.